9A Lesson 4

# DNA, chromosomes and genes

Year 9 Science · 9A Genetics and evolution · Student Book pages 12 to 13

Name _____ Class _____ Date _____

**Today I can:**

☐ I can outline how the structure of DNA was discovered and say what each scientist did.

☐ I can describe how DNA, chromosomes, genes and genetic information are linked, and give the number of chromosomes in body cells and gametes.

**Key words**

**DNA** the very long molecule that stores genetic information

**double helix** the shape of a DNA molecule: two strands twisted into a spiral

**chromosome** a structure in the nucleus made of one long DNA molecule

**gene** a section of DNA that carries the instructions for one characteristic

**genetic information** the instructions for inherited characteristics

**nucleus** the part of a cell that contains the chromosomes

**sex chromosomes** the pair of chromosomes that decides sex: XX (female) or XY (male)

**gamete** a sex cell (sperm or egg) with 23 chromosomes

**zygote** the fertilised egg cell, with 46 chromosomes

**X-ray** a type of radiation used to make images, including of molecules

**model** a simpler representation of something that scientists test against evidence

## ● Core Everyone does this section.

**1.** Match each scientist to one thing they did to help discover the structure of DNA. Draw a line.

| | |
|---|---|
| Francis Crick | took very clear X-ray photos of DNA and said the first model was wrong |
| Rosalind Franklin | tested the model and made corrections to it over several years |
| James Watson | saw from a photo that DNA was a spiral and helped build the model |
| Maurice Wilkins | helped to build the double helix model in 1953 |

**2.** Answer each question by writing a number.

**a)** The number of DNA molecules in one chromosome.

_____

**b)** The number of Y chromosomes in a body cell from a male.

_____

**c)** The number of copies of each gene in a body cell from a female.

_____

**d)** The number of chromosomes in a human gamete.

_____

**e)** The number of chromosomes in a zygote.

_____

**3.** Number these items in order of size. Give the biggest a 1 and the smallest a 4.

____ cell ____ chromosome ____ gene ____ nucleus

**4.** The photo shows the chromosomes from one human body cell, sorted into pairs.

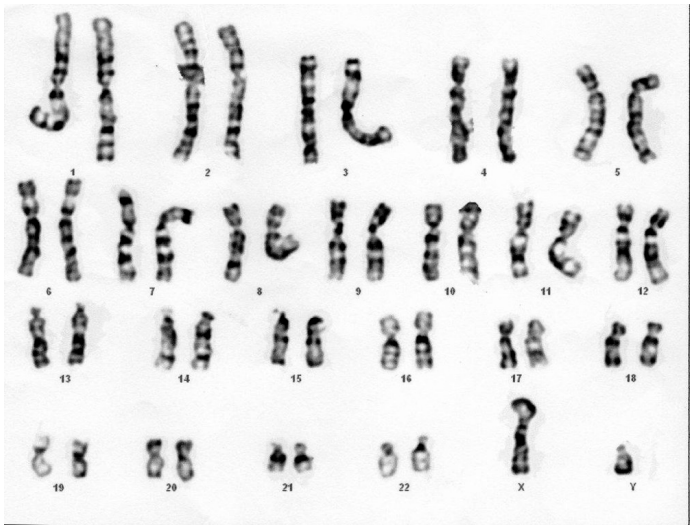



Image: Can H., CC BY 2.0, via Wikimedia Commons

**a)** State whether the cell is from a male or a female. Give a reason.

**b)** How many pairs of chromosomes does the cell contain?

## ●● Stretch When Core is finished.

**S1.** The table shows the number of chromosomes in the body cells of some animals.

**a)** Plot the data on the grid as a bar chart. Put the animals in a logical order and label both axes.

| Animal | lion | elephant | chicken | rabbit | horse |
|---|---|---|---|---|---|
| Number of chromosomes | 38 | 56 | 78 | 44 | 64 |

**b)** Give the reason why the number of chromosomes in each species is an even number.

**c)** A student says: 'This data shows that birds have more chromosomes in their cells than mammals.' Explain whether this is a good conclusion.

**S2.** Explain why a body cell contains two copies of each gene, but a gamete contains only one copy.

## ●●● Challenge Explain in full sentences.

**C1.** Draw a labelled diagram to show how DNA, chromosomes, genes, genetic information and the nucleus are linked. Use arrows or nested shapes.

**C2.** Watson and Crick used data from other scientists to build their model.

**a)** Suggest how they were able to find other scientists' data in 1953.

**b)** Explain why scientists publish their work in journals such as Nature, and why it mattered that Franklin and Wilkins published in the same edition.

## My thinking (any language)

Franklin's X-ray photo was shown to Watson without her permission. Was that fair? What rules should scientists follow when sharing data today?

**Word bank** DNA · double helix · chromosome · gene · genetic information · nucleus · sex chromosomes · gamete · zygote · X-ray

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**Exit ticket 9A Lesson 4: DNA, chromosomes and genes** Name _____

**1.** Where in a cell is DNA found?

_____

**2.** How many chromosomes are there in a human body cell, and in a human gamete?

_____

**3.** What is a gene?

_____

**Today I can:** Goal 1 ☐ Goal 2 ☐